ECE CAPSTONE PROJECT

SMART HOME IoT

Room Temperature & Humidity Monitor

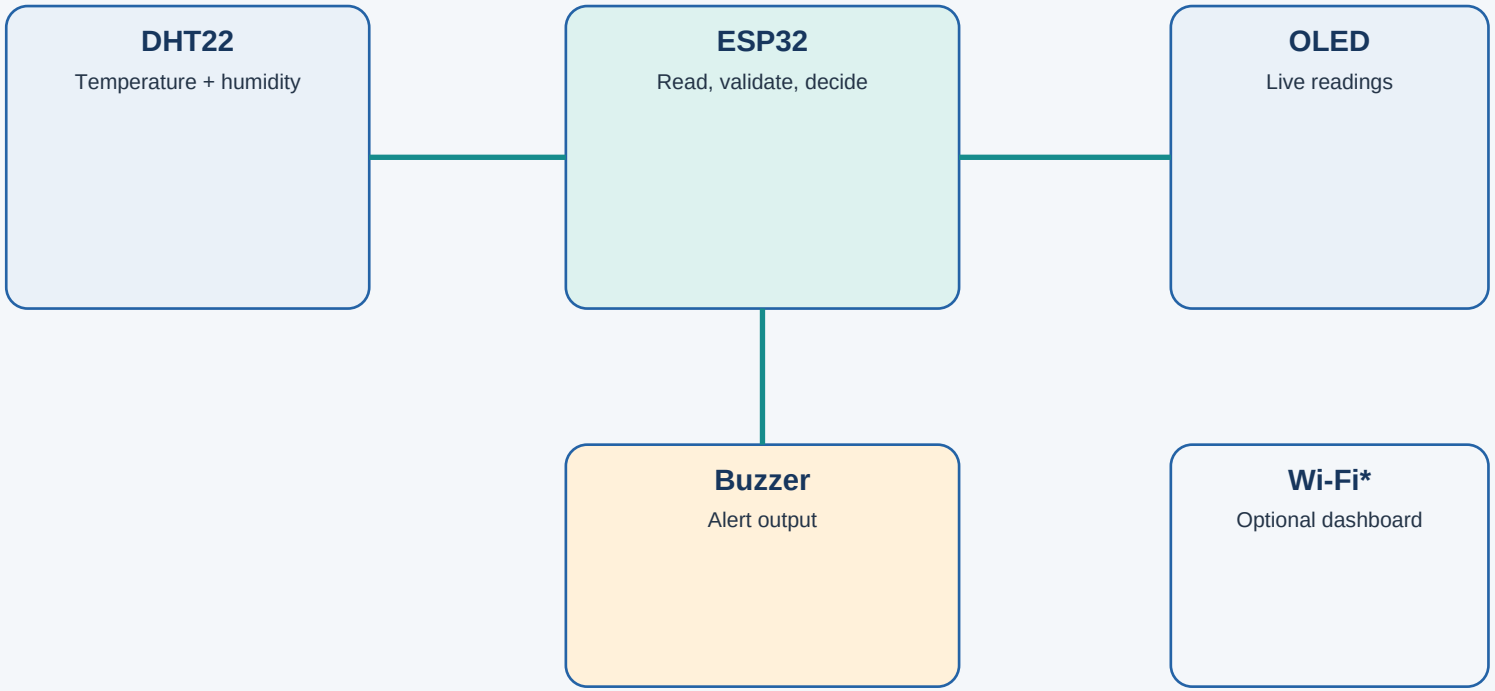
ESP32 • DHT22 • OLED • Local Alert

Student: _____ Register No.: _____

Problem & objectives

- Room conditions are often checked manually and occasionally.
- Measure temperature (°C) and relative humidity (%RH).
- Show live readings on a local OLED screen.
- Sound a buzzer when configurable thresholds are crossed.
- Keep core monitoring operational without Wi-Fi.

System architecture



**Optional network extension; local display and alert do not require cloud access.*

Hardware & estimated BOM

- ESP32 DevKit — controller with integrated Wi-Fi.
- DHT22 / AM2302 — digital temperature and humidity sensor.
- 0.96-inch SSD1306 I²C OLED — local user interface.
- Active buzzer, breadboard, jumper wires and regulated USB supply.
- Indicative total: ■940–■1,900; verify current local quotations.

Wiring & firmware logic

- DHT22 DATA → GPIO 4; VCC → 3V3; GND → GND.
- OLED SDA → GPIO 21; SCL → GPIO 22; VCC → 3V3; GND → GND.
- Buzzer signal → GPIO 18 (use driver if required).
- Sample every ~2 seconds; reject invalid sensor readings.
- Alert if temperature > 30°C OR humidity > 70%RH.

Simulation & verification

- Build circuit in Wokwi or equivalent ESP32 simulator.
- Normal: 24°C / 50%RH → NORMAL, buzzer OFF.
- High temperature: 32°C / 50%RH → ALERT, buzzer ON.
- High humidity: 25°C / 75%RH → ALERT, buzzer ON.
- Test boundaries, sensor error and operation without Wi-Fi.
- Attach genuine screenshots and record observed results.

Safety, privacy & limitations

- Use low-voltage USB power only; never connect mains to breadboard.
- ESP32 GPIO is 3.3 V logic; avoid 5 V signals and excess current.
- Keep Wi-Fi credentials private; restrict dashboard access if added.
- DHT22 accuracy and response are limited; placement affects readings.
- This prototype is not a certified life-safety alarm.

Conclusion & future scope

- Compact embedded IoT monitor with local display and threshold alert.
- Core functions are independent of internet connectivity.
- Next: verify hardware, capture screenshots and complete test log.
- Future: BME280, MQTT/Home Assistant, logging, mobile alerts, enclosure.
- Thank you — Questions?